



December 2025

asking those who feel it

**local and Indigenous knowledge (ILK)
on climate change**

Day 1 (5 December)

Jakob Steiner

Global Awareness Education (GAE)

Transdisciplinary Course Program and Career Service – TRACS
University of Tübingen



@Angharad Dean

- Courses on global issues
- Globalization, human rights, indigenous knowledge, climate crisis exit, etc.
- International team of experts
- **Global Awareness Certificate** for 15 ECTS

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<https://uni-tuebingen.de/en/215290>

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- Alliance of 11 European universities
- GAE participates in many CIVIS projects!
- In this course, there are students who are registered in the **CIVIS micro-program Global Awareness!**



Read more:

<https://civis.eu/en>

Global Awareness Education in CIVIS



- **CIVIS Micro-Program Global Awareness**
 - U of Tübingen, U of Bucharest, and Universidad Autónoma de Madrid
 - 15 ECTS for Bachelor students from all fields, creditable as key qualifications for students from U of Tübingen.



Dr. Inka Koch

Global Awareness Educator & Glaciologist

EBERHARD KARLS
UNIVERSITÄT
TÜBINGEN



WE
VALUE **EXCELLENCE.**

Inka.koch@uni-tuebingen.de

Jakob Steiner - (academic) background

BSc (ETH Zurich, Switzerland): Water for agriculture in Xinjiang/Western China , Prof. Wolfgang Kinzelbach

MSc (ETH Zurich, Switzerland): Ice cliffs on debris-covered glaciers in Nepal , Prof. Paulo Burlando

Concert Diploma Violoncello (Austria/Germany), Prof. Michael Hell

PhD (Univ. Utrecht, Netherlands): Debris-covered glaciers in the high mountain water cycle, Prof. Walter Immerzeel, Prof. Marc Bierkens

Engineer at hydrosolutions ltd. (Switzerland)

- agricultural water management in Central Asia, groundwater monitoring in Eastern and Central China (SDC)

Glacier-Hydrologist at ICIMOD (Nepal, Pakistan)

- group lead of the cryosphere group within the Water and Air Theme, fieldwork and research (Nepal, Pakistan, Afghanistan), trainings for regional member countries, coordination of multi-stakeholder research and policy projects (Nepal, Pakistan, Afghanistan)

(academic) background – current positions

PostDoc at University of Graz (Austria)

– research on the Greenland ice margin



LATTICE - <https://www.uibk.ac.at/en/acinn/research/ice-and-climate/projects/lattice/>

Fellow at the Asian Mountain Academic Alliance (<https://amaa.earth/>) – (co)lead of working groups on cryosphere, society and data

<https://fidelsteiner.github.io> / jakob.steiner@uni-graz.at



interdisciplinary transgressions



#IMC: September 14 - 18 2025



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Approaching cold mountains through local knowledge exchange and collaboration



♥ Add Session to my Favorites

#IMC25 > FS 3.170: Diversifying expertise on cold mountain regions – Decentering scientific epistemologies through local knowledge and collaboration on eye level

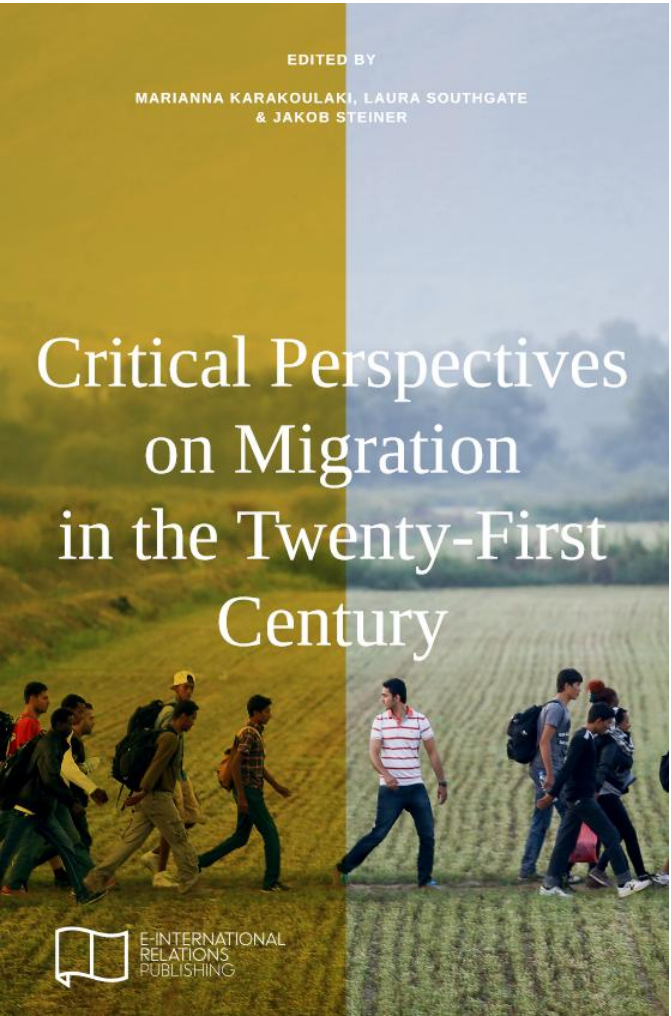
Details	Description
 Full Title FS 3.170: Diversifying expertise on cold mountain regions - Decentering scientific epistemologies through local knowledge and collaboration on eye level	Nothing captures common attention for anthropogenic global warming like the changing cryosphere: glaciers are receding, permafrost is thawing, snow in winter is becoming less reliable, avalanche patterns are shifting. To understand the socioecological implications of these changes, there is a need to listen to the people who have lived with snow and ice for centuries. Their intimate knowledge adds cosmological, sensory and practical dimensions to the technoscientific approaches of glaciologists. Local communities are the ones who have to adapt to changing scenarios, and often do so in ingenious ways. Exchange with them – or the absence of it – influences the way sci-
 Scheduled TBA	
 Convener Anna-Maria Walter	

JAKOB STEINER

NEGLECTED LAND IN BETWEEN – THE OBLIVIOUSNESS TO SMALL TOWN PAKISTAN IN THE WESTERN NARRATIVE

Abstract

The Western imagination of human settlement of the Subcontinent has long been limited to two extreme possibilities - the backward but romantic village in the remote countryside versus the bustling megacity. Ideally, as the representation of the ‘modern’ Pakistan, India or Bangladesh these worlds collide in literary, cinematic or even scholarly representation. The world in between, life in the small town, the urban peripheries of larger conglomerations and the exchanges between these spaces and their influence on urban growth and rural development have been largely neglected. Not only has this resulted in a very biased imagination of these countries but also in inadequate approaches in infrastructure development funded and thought up outside the country and by the NGO sector that often tries to emulate the western imagination. This paper examines some of these deficiencies. It begins with discussing briefly how Pakistan is framed by the foreign eye that had a chance to actually see the place and where these frames likely originate. It is the simple example of travelling between Lahore and Islamabad, a route often taken by decision makers in international organisations and policy wonks that holds as an initial introduction to the dilemma. The route mostly taken, the Motorway, reproduces the bimodal view of Pakistan of megacity vs. rural village. On the other hand, the GT road, which leads one through the whole spectrum of urban settlements in the country, from the historical metropolis Lahore through the largely unmentioned cities in narratives on Pakistan such as, Gujranwala, Jhelum or Wazirabad to the twin cities, with many smaller towns on the way, is hardly known.



Welcome to the class 😊

Total of ~15 students

Field of studies:

- Biology
- Bioengineering
- Computer Linguistics
- Education
- Political Science
- Social Anthropology

Course content

5 December (1400 – 1800)

Pre course evaluation - general introduction and history – guest lecture (Arctic)- introduction to writing assignment

6 December (0900 – 1700)

Guest lecture (Himalaya) – Climate Science– interactive team work - International efforts to integrate ILK

7 December (0900 – 1700)

ILK governance and ethics – integrating ILK with conventional data - interactive team work - group presentation and discussion

to prepare

- written assignments to be introduced today
- a final presentation:
 - form a group of 2 to 4
 - choose a study/topic to dissect and analyse their use of Indigenous and local knowledge, discussing what you learned during the course
 - present in a short 10 min presentation on the 7th of December
- 2 guest lectures – read up on their background to be prepared with questions!

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5 December/ Day 1

Session 1 (5 December) ✎

Sammelaktionen ✎



Welcome to our first session 😊

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Self evaluation

https://docs.google.com/forms/d/e/1FAIpQLSfUNdBNEySHnr1Mat3ROI3gqPSXI2sPXf_2JRo_nNHeF5nlZA/viewform?usp=dialog



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Questions regarding the assignments

Assignment 1 and 2 should be straightforward

Assignment 3

Choose one of the publications attached here and prepare a short (max. 1 page) essay, summarizing at least one of the core principles and providing a discussion of lines of argument you think can be challenged, especially for the case of publications dating back to the 1990s.

Some tips:

- Scan abstract/conclusion to see which article most likely appeals to you. If you know of another suitable paper you would want to discuss, please go ahead – it has to stay on topic!
- You may read more than one paper of course – but settle on one central argument made.
- You may use Google/ChatGPT/etc to search about certain concepts and are encouraged to refer to other literature – but write an independent text and use proper referencing!
- You are encouraged to refer to your own experience. When presenting your own opinion argue for it coherently.

Final presentation

In this course you learned about the role of Indigenous and Local Knowledge (ILK) in climate research and the current shortcomings as well as how difficult it can be to grasp concepts associated with this topic. We now encourage you to choose a research study or a project that includes such knowledge in its approach. You should present the concept behind the work, discuss if and how in your view ILK was included successfully and whether it was documented appropriately. We encourage you to bring in your own views on how you would have done things differently and why.

You are encouraged to form groups of 3 to 4 students to present. Choose a research study or professional project (e.g., from international development or a project report that is not necessarily published in a scientific journal) that has Indigenous and/or local knowledge at its core. You are encouraged to choose an example from your own background. It does not necessarily need to be related to climate research or the Arctic or mountain environments. What is more important is that it is a topic you are motivated to investigate and can evaluate given the resources you have access to.

You are expected to present your findings in a 10-minute presentation, where everyone has an equal share of presenting. Ideally you do this in a slideshow format. However, you are free to choose the final format. Please see the evaluation rubric that we will use to assess your work (shown below as a PDF file).

Course aims

At the end of the course, you will be able to assess

- (i) why and when Indigenous and local knowledge is crucial,
- (ii) in which international frameworks it is being proposed as an important aspect of understanding and better adapting to climate change and
- (iii) what the main challenges are to integrate it with other types of knowledge.

What to expect from the course

- to understand what local and Indigenous knowledge is and critically reflect on its use in climate change related research
- to learn from examples of its application on how this knowledge is crucial but also how it can be misused

What to bring to the course

- personal reflection on knowledge from your own background
- challenge the status quo – this isn't a topic that is set in stone – your opinion and knowledge matters and your contribution can shape the discourse!

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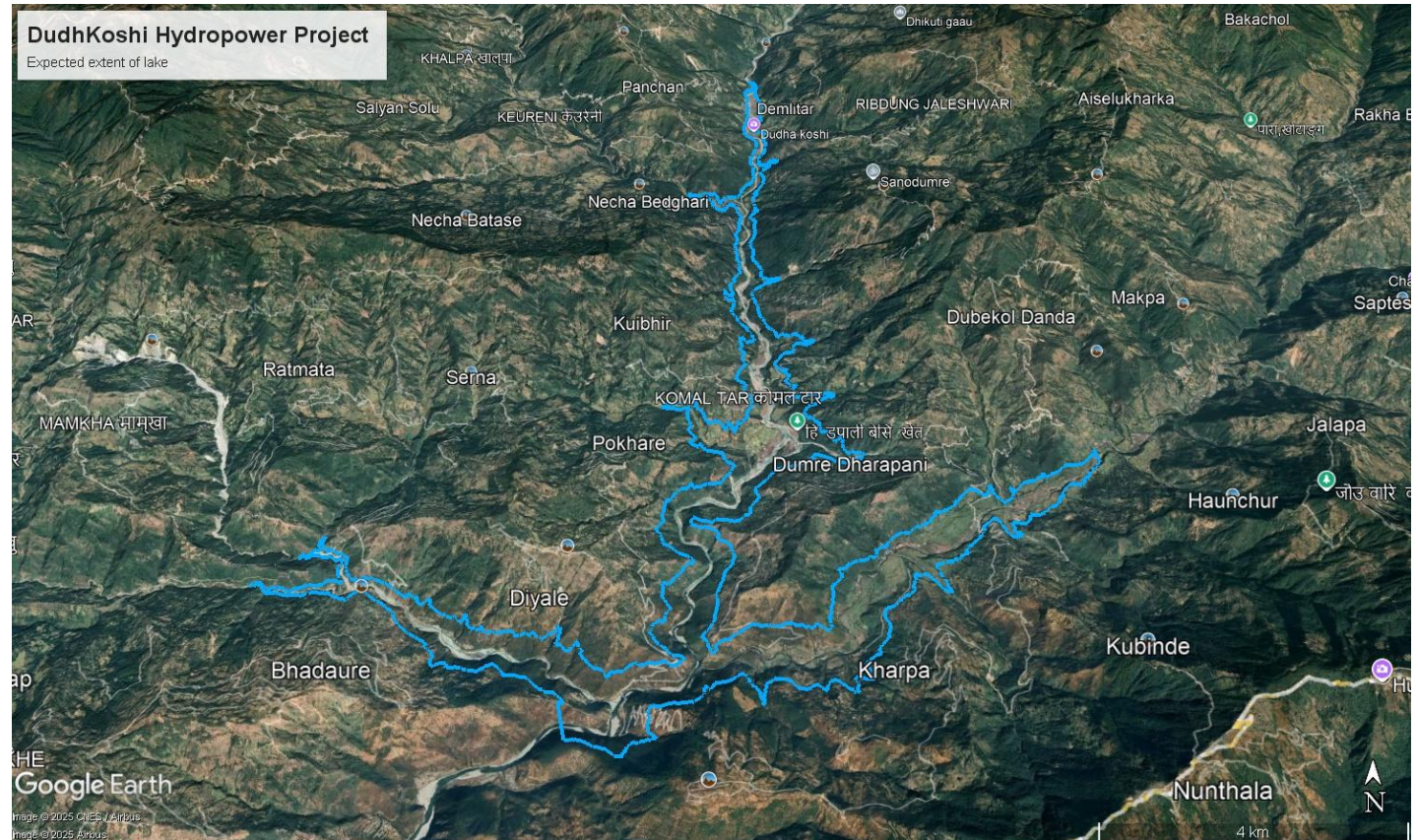
Challenge for this course

- 1) Think about the concept of averages and extremes in our perception and how they define our choices, planning, capacity to adapt to potentially future adverse effects.

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FORUM-ASIA's Post

FORUM-ASIA
May 2 · 🌐

#Nepal's push for clean energy through the ADB-funded #Dudhkoshi Hydropower Project may seem like progress—but for the #Majhi Indigenous community in Rawabesi, it brings fear of displacement, environmental harm, and the loss of cultural identity. Development that harms communities cannot be called inclusive.

The project threatens to alter the flow of the Dudhkoshi River, a lifeline for the Majhi people. For generations, they have relied on the river not just for their livelihood, but as a core part of who they are. As bulldozers and blueprints arrive, their voices are being sidelined in the name of development.

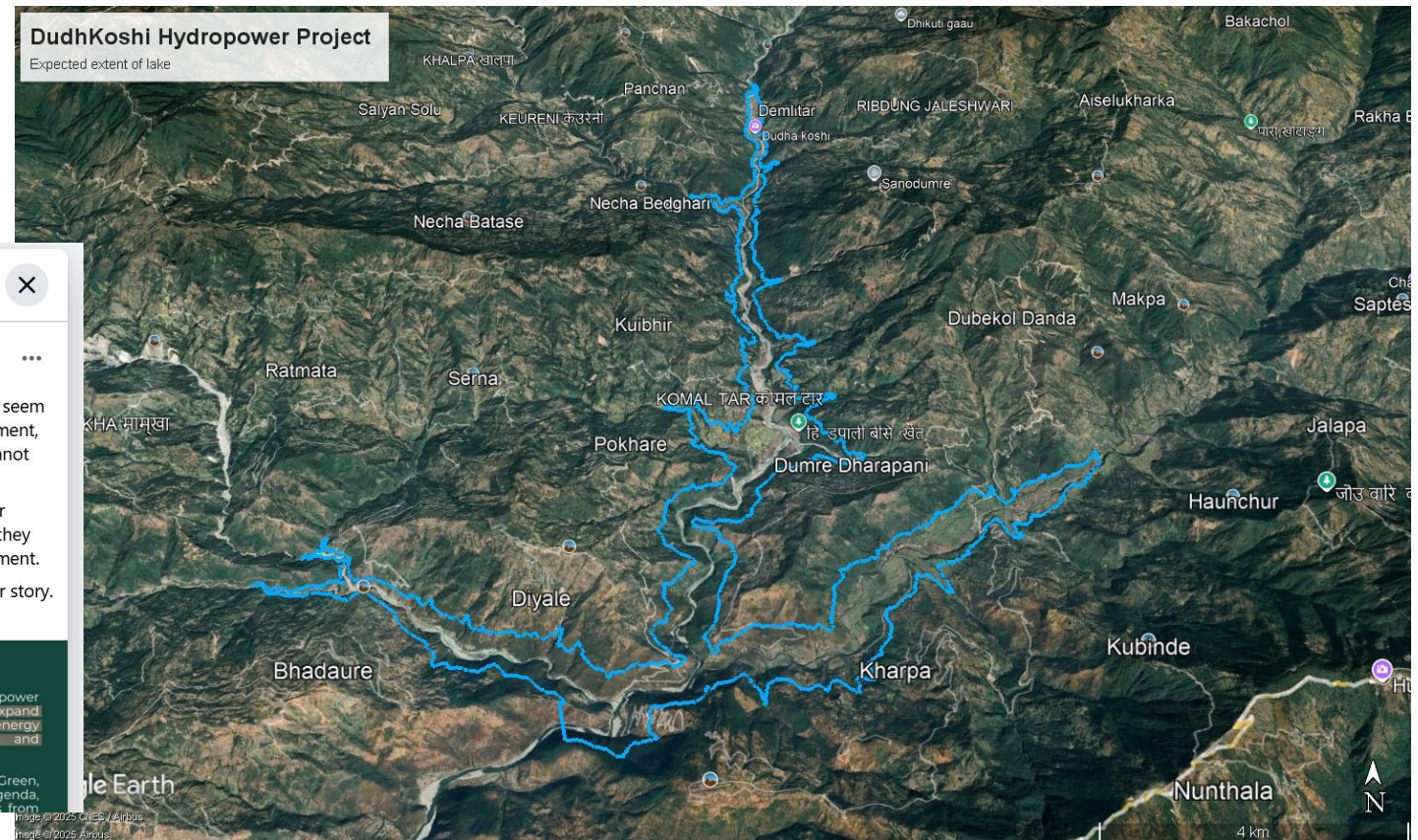
👂 Stay tuned for our video series, Hear Me Out, where a Majhi community member shares their story. It's time to listen—to protect both people and the planet.

Photo Source: The Independent & Cultural Survival

FORUM-ASIA

The ADB is financing the Dudhkoshi Hydropower Project in collaboration with the World Bank to expand Nepal's clean energy supply, boost regional energy trade and contribute to climate resilience and sustainable development.

The project is part of Nepal's 2.2 billion USD Green, Resilient, and Inclusive Development (GRID) agenda, largely financed by the ADB, with contributions from



General definitions

Local knowledge:

Knowledge that anyone associated to a certain geographical and temporal space can acquire and use.

e.g. I hold local knowledge with respect to Pakistan.

Indigenous knowledge:

Knowledge passed down through generations, acquired and applied by a society; historically and culturally enriched and changing but persistent.

e.g. I can not hold Indigenous knowledge with respect to Pakistan.

General definitions

UNESCO (<https://www.unesco.org/en/links>):

*Local and indigenous knowledge refers to the **understandings, skills and philosophies developed by societies with long histories of interaction with their natural surroundings**. For rural and indigenous peoples, local knowledge **informs decision-making** about fundamental aspects of day-to-day life.*

IPBES (<https://www.ipbes.net/indigenous-local-knowledge>):

*IPBES **recognizes and respects** the contributions of Indigenous and local knowledge to the conservation and sustainable use of biodiversity. The IPBES Global Assessment (2019) showed that at least a quarter of the global land area is traditionally owned, managed, used or occupied by **Indigenous Peoples**, and that in addition, a diverse array of **local communities** manage significant areas.*

General definitions

IPCC (<https://www.ipcc.ch/srccl/faqs/faqs-chapter-7/>):

Indigenous knowledge (IK) refers to the **understandings, skills and philosophies developed by societies with long histories of interaction with their natural surroundings**. Local knowledge (LK) refers to the **understandings and skills developed by individuals and populations, specific to the place where they live**. These forms of knowledge, jointly referred to as Indigenous and Local Knowledge or ILK, are often highly context specific and embedded in local institutions, providing biological and ecosystem knowledge with landscape information.

Am I a local/Indigenous knowledge holder?



Many (confusing) terminologies

In the course for sake or space we'll often stick to ILK
(Indigenous and local knowledge)

Antweiler (1998)

Table 1: Diversity of terms for local knowledge (and its branches) and their various connotations (compiled from the literature of the 1960s to 1997).

Term, synonyms	meaning, salient aspect, implicit significance, antonym
• indigenous knowledge (internationally the most widespread term)	culturally integrated knowledge; knowledge of small, marginal/non-Western groups
• endogenous knowledge	of internal origin, as opposed to exogenous or external knowledge
• native knowledge/expertise	implies knowledge of a natural character, closeness to nature
• local knowledge	knowledge rooted in local or regional culture and ecology
• sustainable knowledge	sustainable within the natural and cultural environment
• traditional knowledge	handed down, old, oral (implying static, low level of change)
• autochthonous knowledge	of internal origin, culturally integrated
• people's knowledge	broadly disseminated knowledge, knowledge as potential for political resistance, as opposed to elite knowledge
• folk knowledge, folk science, folk competence	traditional, rural (in industrial societies)
• little tradition	tends to denote oral knowledge, as opposed to great tradition
• community knowledge	related to small social units
• cultural knowledge, cognition (in the restricted sense)	culturally integrated and practice-oriented
• ethnic knowledge	related to an ethnic "we"-group (ethnicity)
• culturally specific knowledge	specificity, singularity, particularity
• ethnoscience (used here to denote local knowledge; previously used to denote the field of research!)	scientific (systematic) character; examples are: ethnobotany, ethnosociology, ethnomedicine, ethnopharmacology, ethnoepidemiology
• (cultural) knowledge system (superseded "ethnoscience")	systematic character, generating rules (if x then y) and structures
• (cultural) belief system, (cultural) meaning system	means the same as "knowledge system," but implies a less scientific character
• everyday knowledge, practical knowledge, mundane cognition, vernacular, common sense, generalist	informal, practical, applied, as opposed to academic, specialist, expert knowledge or as opposed to ritual knowledge
• science of the concrete	based on that which actually exists/is visible
• experiential knowledge	as opposed to theoretical knowledge, speculation
• experimental knowledge	trial and error, as opposed to controlled experiment
• farmers' knowledge	knowledge relating to the farm as an economic unit
• peasant knowledge	as opposed to elite knowledge; implies experiences of dependency

What is (Indigenous and local) knowledge?

1. Environmental knowledge or cognition
 1. Knowledge on the natural environment (plants, ecosystems, natural disasters)
 2. Knowledge on anthropogenically modified environment (risks, management of resources, agricultural knowledge)
 3. Knowledge on the social and political environment (state, community structures, power dynamics, development projects)
2. Medical knowledge
3. Indigenous technical knowledge
4. Organisational and management knowledge (including legal knowledge)
5. Knowledge on persons, relationships (social cognition)

What is (Indigenous and local) knowledge?

1. WHAT - How many avalanches have happened in your valley in the last 50 years?
2. HOW - How does the local community deal with avalanche risk and related mitigation and adaptation?

What is (Indigenous and local) knowledge?

1. Declarative knowledge
 1. Factual
 2. Categorical (kinship, colours ...)
2. Procedural knowledge
 1. General processes (e.g. farming calendar, development project cycle)
 2. Specific processes (e.g. how to greet others in a community, how to enter a restaurant, how to tip a waiter)
3. Complex knowledge (belief systems, societal norms, decision making procedures)

Knowledge isn't static

When assessing we need to take stock of

- the present knowledge pool (*all that is currently known*)
- the present spatial and social distribution (*who holds knowledge*)
- the acquisition process of knowledge (*acquisition or diffusion*)
- the social transmission of knowledge



Khaplu, Baltistan
Pakistan, July 2023



Khaplu, Baltistan
Pakistan, July 2023



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Shigar, Baltistan
Pakistan, July 2023



Skardu, Baltistan
July 2023

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Kyanjing, Rasuwa
Nepal, April 2019

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Reasons for *unknowing*

- knowledge can be unknown because it is not available
- knowledge isn't available because it is not understood
- knowledge is not discussed because it is being taken for granted
- actively and consciously hidden knowledge (e.g. political reasons)
- distorted knowledge (again possibly to influence a debate)

the *emic* view – what do 'locals' constitute as 'factual knowledge' vs. what they consider to be 'credibly claimed'



What is (Indigenous and local) knowledge?

Alternative terminology:

- 'local' not referring to a specific point in space but 'knowledge is rooted in a certain 'culture' or 'ecology' – 'knowledge is culturally situated'
- 'knowledge of social systems' – as such empirical, factual and comprehensive but not necessarily conscious, verbalized or susceptible to verbalization/visualization
- knowledge can only be understood in relation to a situation (referential transparency) and is related to practices and situations (situated knowledge)

kind answers questions such as: "How do I enter a house in X?", "How do I order a drink in Y?", or "How do I find a flat in Z?" (examples from Frake 1964). The answers to these apparently simple

Antweiler (1998)

Other terminology

Traditional ecological knowledge (TEK) - *the knowledge and insights acquired through extensive observation of an area or species (Huntington, 2011)*

Indeginismo/indeginista (spanish) – strong political connotation

Other terminology

Traditional ecological knowledge (TEK) - *the knowledge and insights acquired through extensive observation of an area or species (Huntington, 2011)*

Indeginismo/indeginista (spanish) – strong political connotation

Climate – long term (>30 year) regional or global average of weather variables (temperature, humidity, precipitation etc)

Governance - The way in which a system is managed, or resources are allocated, and the processes of influencing, steering, controlling or managing the actions of different parts of society to achieve management goals.

Western Science – conventional science (i.e. all that is not ILK) dominated in history by the 'West' or 'Global North'

Climate (change) science

Climate science (climatology): interdisciplinary field including meteorology, geography, geology, oceanography and physics



hydrology,
geomorphology,
cryosphere

Climate change research: intradisciplinary field reaching into social sciences, economics, humanities ...

International body in climate change research



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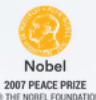
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The Intergovernmental Panel on Climate Change

The Intergovernmental Panel on Climate Change (IPCC) is the United Nations body for assessing the science related to climate change.

[IPCC-60](#)

[VACANCIES](#)

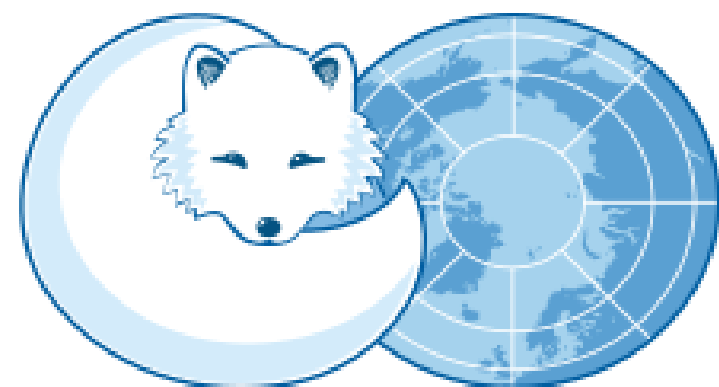


Beyond the strict climate science



Science and Policy
for People and Nature

ICIMOD



ARCTIC COUNCIL



IASC

Disaster Risk Reduction (DRR)

Disaster Risk Reduction is the concept and practice of reducing disaster risks through systematic efforts to analyse and manage the causal factors of disasters (UNDRR)

Loss and damage caused by climate change has already uprooted the lives and livelihoods of millions of people, and extreme weather events are endangering communities everywhere. One-third of Pakistan has been submerged. Pacific island nations are on the brink of extinction due to rising water levels.

In 2020 alone, 980 natural disasters cost the global economy more than US\$210 billion.

The future outlook is even more serious.

Without action on climate risk reduction, it is projected that by 2030 there will be more than one disaster every day.

According to a recent UNDP study, if emissions reductions remain moderate, climate change could kill more than 40 million people by the end of the century.

Asako Okai, ASG UNDP, March 2023 - <https://www.undp.org/speeches/directions-disaster-risk-reduction-era-climate-change>

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Nicole Gaul

Status: MSc Candidate

Program: MSc in Physical Geography

Supervisor(s): Dr. Robert Way

Time period: 2024-

Funding stream: CRC & CINUK

Project title: Exploring glacier change through youth-engaged participatory glaciology in Tongait KakKasuangita SilakKijapvinga (Torngat Mountains National Park, Nunatsiavut)



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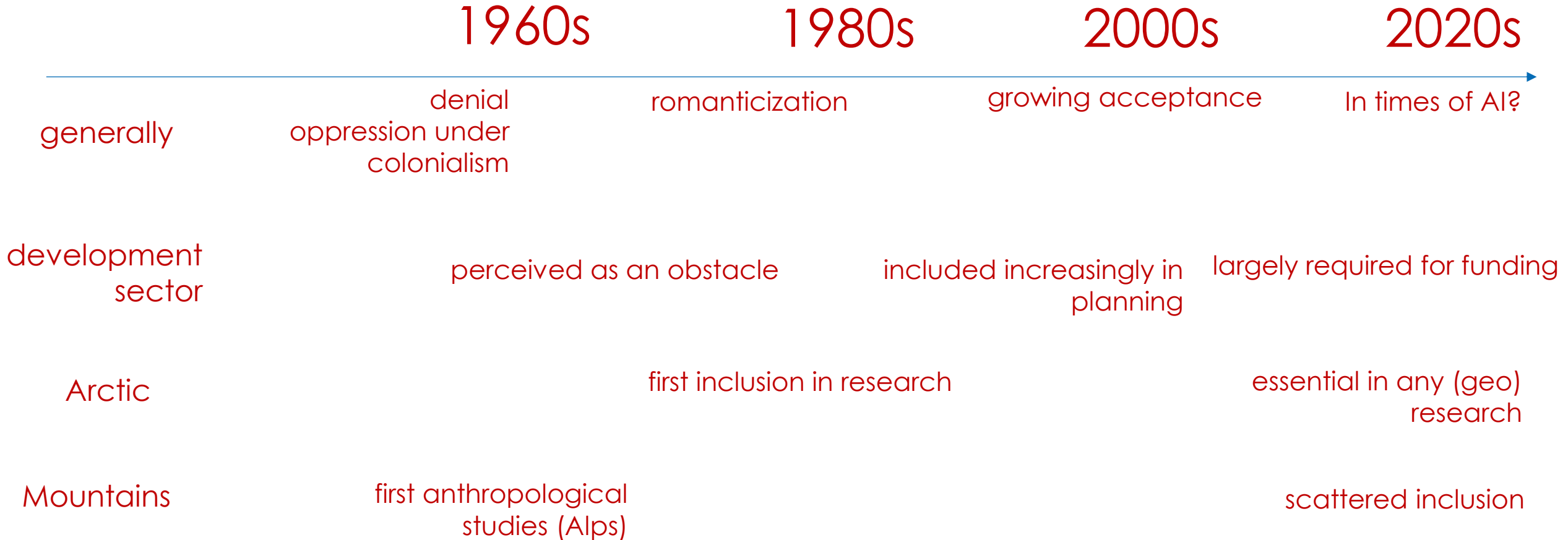
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History of ILK in general



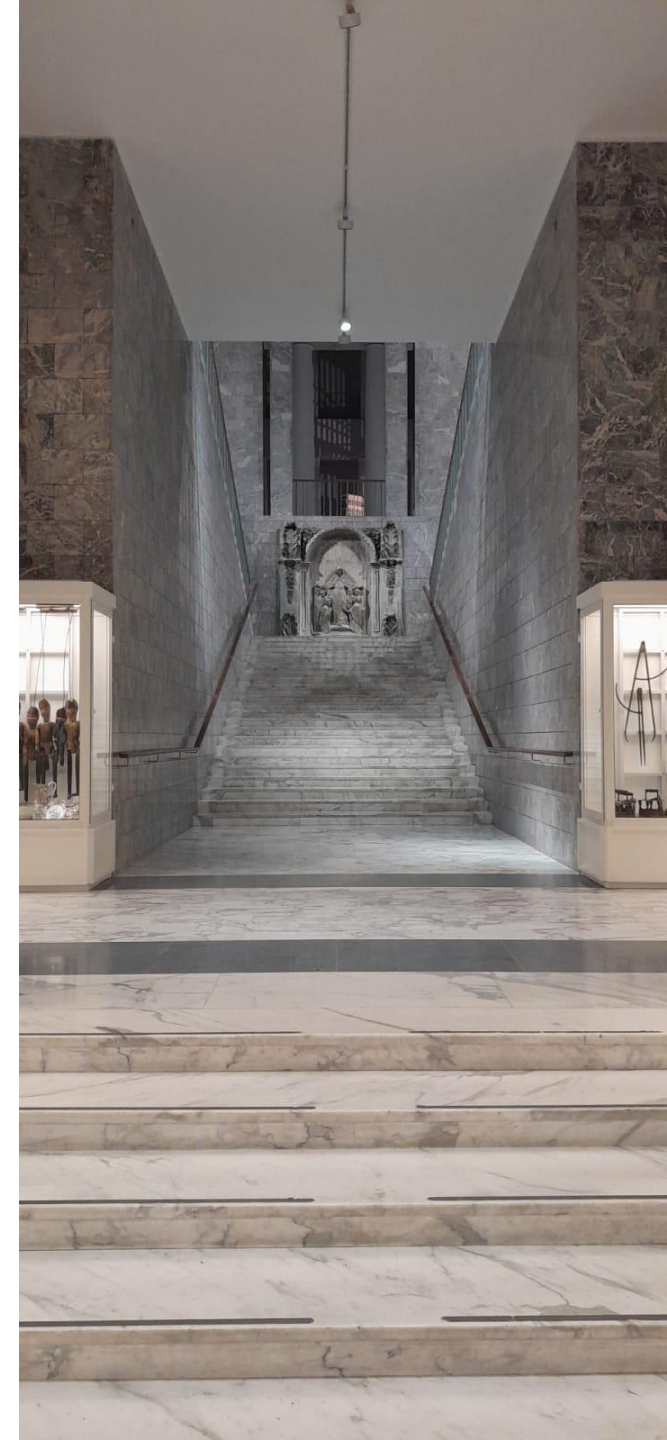
ILK in colonial pasts (and presents)

- it is 'useful' for colonialists/rulers/scientists to document definitely
- ...causing the value of oral histories to diminish
- ... corrupting the process of collecting this knowledge (i.e. inventing stories that we want to hear)

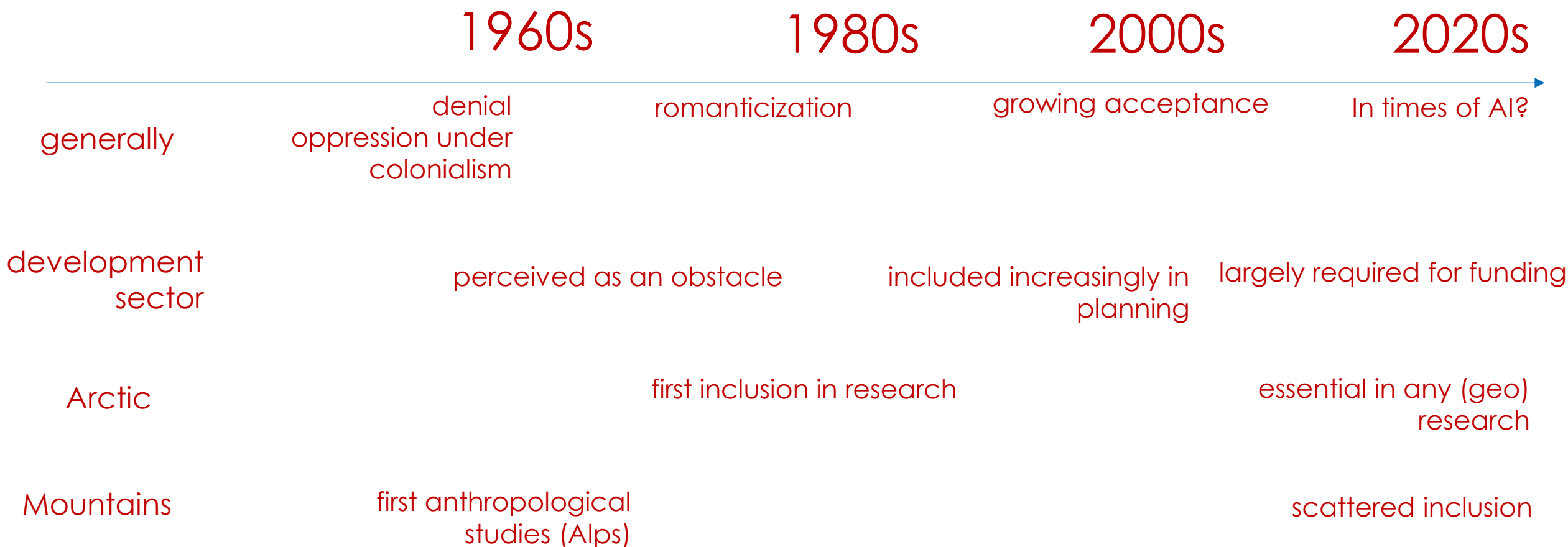
Introduction: popular memory

Popular memory is memory which lives; memory that survives orally, transferred between generations. It is inherently connected to family and community and is experienced subjectively beyond the realm of linear and continuous development. Popular memory uses language to mediate between past and present and grants substance to historical imagination. Simultaneously individual, familial, and collective, popular memory is often at risk of being diluted or fragmented through the institutional process of documentation, archiving, museum preservation, commemorative celebration, and other material methods used to coagulate the immaterial culture of the past into present cultural memory. Alienated and abstracted from their living contexts, the objects displayed here use the museum showcase as a means of transmission, attempting also to counter the fate of digital oblivion.

Museo
Nazionale delle Arti
e Tradizioni Popolari
Rome, January 2024



History of ILK in general



History of ILK in general

- *putting people first* (Cernea, 1991) – South vs North (West) concept already mentioned
- local knowledge allows for measures which begin where people are - not 'where others want them to be' (Marsden, 1994)
- local self reliance and direct participatory democracy, starting point the locality because civil society is most readily mobilized around local issues (Friedman, 1992)

but is ILK always 'good' or 'the solution'?

- local knowledge can be targeted at (short term) survival, which may contribute to overexploitation of local resources -> **integration with exogenous information**
- local knowledge can be used by external actors to control local communities -> **ethics**
- knowledge in a community is hardly ever equally distributed -> **politics**

Integration with endogenous knowledge forms

local knowledge. It is to define to which extent external knowledge (and information) needs to be introduced and integrated into local knowledge systems as a constructive contribution to local adaptation to a changing environment.

Antweiler (1998)

With regard to knowledge, rhetoric, and behaviour, we are presently confronted with a situation in many communities that does not allow the application of existing knowledge. Knowledge and rhetoric – equally in the countries of the North and the South – do not match the current unsustainable behaviour. Due to the fact that knowledge gained through oral traditions, information flow of media, formal training, or experience is not longer fully and systematically used, we have to deal with a “fragmented” local knowledge system (Mersmann 1993). This implies that the knowledge on social, cultural, and particularly ecological details, often fascinating in their abundance, do not practically add to locally adapted paths of endogenous, self-reliant development and the sustainable use of natural resources. Another consequence of this situation are the difficulties of local communities and individuals to actively integrate external information, experience, and knowledge into the local knowledge system.

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ILK is political

- (local/indigenous) knowledge is not just a 'descriptive term' but has political connotation (Antweiler, 1998; Foucault, 1970)
- problematic assumption that 'knowledge is evenly distributed' – knowledge holders and hence 'users for power' exist at every level – a prerequisite to rely on local knowledge is hence 'intact cultural and social context'

The process of acquiring insight produces awareness of one's own way of life as something worthwhile, and the mastery of skills breeds self-esteem. Thus, knowledge is created in a combination of utilitarian and symbolic practice – not in a vacuum: A plant is considered edible or characterized as just grass or weed; it is important that I know the difference, and it may turn out convenient to *demonstrate* that I possess the knowledge.

(Siverts 1991: 308) Antweiler (1998)

History of ILK in general

'intact cultural and social context' (Antweiler, 1998)

- how is that possible in communities under pressure (from climate change) and faced with a lot of attention (with respect to climate change)?
- examples High Mountain Asia and Greenland



<https://www.theguardian.com/world/2025/jan/10/greenland-views-donald-trump-advances>



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Thank you

